

MIGUEL SALVACION

msalvacion@cmu.edu | miguelsalv.com | linkedin.com/in/msalvacion | github.com/miguel-salv

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. and M.S. in Electrical and Computer Engineering — GPA: 3.81/4.00 — Dean's List

May 2027

Leadership: Filipino Student Association Treasurer – Managed \$9,000 budget for 150+ member organization

TECHNICAL SKILLS

Languages: C/C++, ARM Assembly (ARMv7-M Thumb-2), Python, SystemVerilog, MATLAB

Protocols & Interfaces: CAN, I2C, SPI, UART, PWM, ADC, DMA

Embedded & RTOS: Zephyr RTOS, STM32 (Cortex-M4), Teensy 4.1, Linux, GDB, OpenOCD, JTAG

Hardware & EDA: KiCad, Cadence Virtuoso, Vivado, FPGA, SolidWorks, Autodesk Fusion, PCB Design

WORK EXPERIENCE

Hardware Systems Engineering Intern | *eDrive Units, Diagnostics*

Summer 2025 & 2026

Ford Motor Company

Detroit, MI

- Resolved a 5.8V drop across an EV control unit via schematic review, identifying elevated temperatures and undersized wiring as root causes and specifying upgraded wire gauges and routing changes.
- Analyzed field warranty data and hardware teardowns across production EV platforms, isolating a mechanical park-lock mechanism fault and presenting an 8D root-cause analysis for engineering leadership.
- Authored secondary electric drive unit teardown and service procedures, establishing a modular eMotor sub-assembly repair path to eliminate full-unit replacements in the field.

Embedded Systems Researcher | *Semiconductor Fab, RF Hardware*

Jan 2026 – Present

CMU Hacker Fab

Pittsburgh, PA

- Designed and built an open-source, automated RF impedance matching network for \$636 (80%+ cost reduction over commercial units), eliminating manual tuning during thin-film depositions.
- Achieved a 1.2 VSWR at 95–100 W and sustained a stable plasma for 60+ minutes by implementing real-time coordinate descent on a Teensy 4.1 driving variable capacitors.
- Designed a custom control PCB in KiCad and verified SWR sensing and power-protection circuits in ngspice, ensuring ADC inputs never exceeded the 3.3V rail across parasitic and component corner runs.

Head Teaching Assistant | *RTOS, Firmware, ARM Architecture*

Jan 2026 – Present

CMU 18-349: Introduction to Embedded Systems

Pittsburgh, PA

- Managed 10 TAs and mentored 60+ students debugging custom preemptive RTOS kernels, resolving PendSV context-switch faults, RMS bugs, and HLP priority-inversion edge cases using GDB and OpenOCD over JTAG.
- Guided bare-metal driver development on STM32 Cortex-M4 targets, covering direct MMIO register configuration, interrupt-driven serial buses (UART/SPI/I2C), and hardware PWM timers.
- Conducted schematic and layout reviews for student 2-layer PCBs, assisting hardware-software integration for closed-loop PID motor control and encoder feedback.

PROJECTS

Drive-by-Wire System (In Progress) | *Zephyr, STM32, CAN*

Aug 2026 – Present

- Developing a three-node system for distributed steering and throttle control using STM32s and Zephyr RTOS.
- Designed redundant CAN messaging and fault-tolerant synchronization targeting below 50 ms end-to-end latency.

Preemptive RTOS & Vehicle Platform | *ARM Assembly, C, RMS, Fusion 360*

Aug 2025 – Dec 2025

- Built a preemptive real-time kernel for STM32 in C and ARM Assembly, featuring PendSV context switching, newlib SVC syscalls, and Highest Locker Protocol mutexes.
- Implemented Rate-Monotonic Scheduling to ensure zero deadline misses across concurrent closed-loop PID speed control, UART telemetry, and I2C LCD tasks.
- Designed a 2-layer PCB in Fusion and wrote bare-metal drivers for quadrature decoding, PWM, UART, and I2C.

Autonomous Trash Collection Robot | *Raspberry Pi, Arduino, UART*

Oct 2024 – Jan 2025

- Built a four-motor robot that achieved 90% bottle-detection accuracy across 20 trials by integrating Raspberry Pi vision with Arduino motor control over UART.